AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 9 · WORKBOOK

Experimental units and independence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete unit, measurement and group-split record.

Keep entity roles, IDs, repeat counts, split membership, assignment design and unverified assumptions together.

Keep one habit: Recall → W1–W3/W5 → independent classroom-unit case → unit/split project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

Baselines and fair comparisons

> **My remembered idea, example and correction**

A literature comparison matrix

> **My remembered idea, example and correction**

Finding primary research

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Name treatment-assignment, sampling and measurement entities separately. An experimental unit receives a treatment in the design; repeated observations do not automatically become more such units. Main: Q1, Q2 and Q3 each have four runs, giving twelve measurements across three query contexts. Preserve pairing and state the actual assignment/sampling assumptions.

  Separate picture: P1 through P4 each have three views, giving twelve photo records from four plant objects. Train on all P1/P2 views and evaluate all P3/P4 views: six per split with no shared plant. Moving P1-v1 to evaluation while other P1 views train gives five training views, seven evaluation views and shared P1, violating the new-plant goal.

  Object identity matters, not merely filename uniqueness or balanced counts. Distinct/disjoint IDs do not establish independence or representative sampling. Four plants provide limited coverage. Native diagram and code use fictional metadata, with no actual images, randomization, training or participant data. Split disjointness is one grouping check, not a performance or broad-reliability result.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Name treatment-assignment, sampling and measurement entities separately. An experimental unit receives a treatment in the design; repeated observations do not automatically become more such units. Main: Q1, Q2 and Q3 each have four runs, giving twelve measurements across three query contexts. Preserve pairing and state the actual assignment/sampling assumptions.

  Separate picture: P1 through P4 each have three views, giving twelve photo records from four plant objects. Train on all P1/P2 views and evaluate all P3/P4 views: six per split with no shared plant. Moving P1-v1 to evaluation while other P1 views train gives five training views, seven evaluation views and shared P1, violating the new-plant goal.

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- Three queries each run four times produce 12 **output**s. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Name treatment-assignment, sampling and measurement entities separately. An experimental unit receives a treatment in the design; repeated observations do not automatically become more such units. Main: Q1, Q2 and Q3 each have four runs, giving twelve measurements across three query contexts. Preserve pairing and state the actual assignment/sampling assumptions.

  Separate picture: P1 through P4 each have three views, giving twelve photo records from four plant objects. Train on all P1/P2 views and evaluate all P3/P4 views: six per split with no shared plant. Moving P1-v1 to evaluation while other P1 views train gives five training views, seven evaluation views and shared P1, violating the new-plant goal.

  Object identity matters, not merely filename uniqueness or balanced counts. Distinct/disjoint IDs do not establish independence or representative sampling. Four plants provide limited coverage. Native diagram and code use fictional metadata, with no actual images, randomization, training or participant data. Split disjointness is one grouping check, not a performance or broad-reliability result.

- A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Name treatment-assignment, sampling and measurement entities separately. An experimental unit receives a treatment in the design; repeated observations do not automatically become more such units. Main: Q1, Q2 and Q3 each have four runs, giving twelve measurements across three query contexts. Preserve pairing and state the actual assignment/sampling assumptions.

  Separate picture: P1 through P4 each have three views, giving twelve photo records from four plant objects. Train on all P1/P2 views and evaluate all P3/P4 views: six per split with no shared plant. Moving P1-v1 to evaluation while other P1 views train gives five training views, seven evaluation views and shared P1, violating the new-plant goal.

  Object identity matters, not merely filename uniqueness or balanced counts. Distinct/disjoint IDs do not establish independence or representative sampling. Four plants provide limited coverage. Native diagram and code use fictional metadata, with no actual images, randomization, training or participant data. Split disjointness is one grouping check, not a performance or broad-reliability result.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **experimental unit**s and independence **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **experimental unit**s and independence.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Experimental unit**s and independence. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Experimental unit**s and independence. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

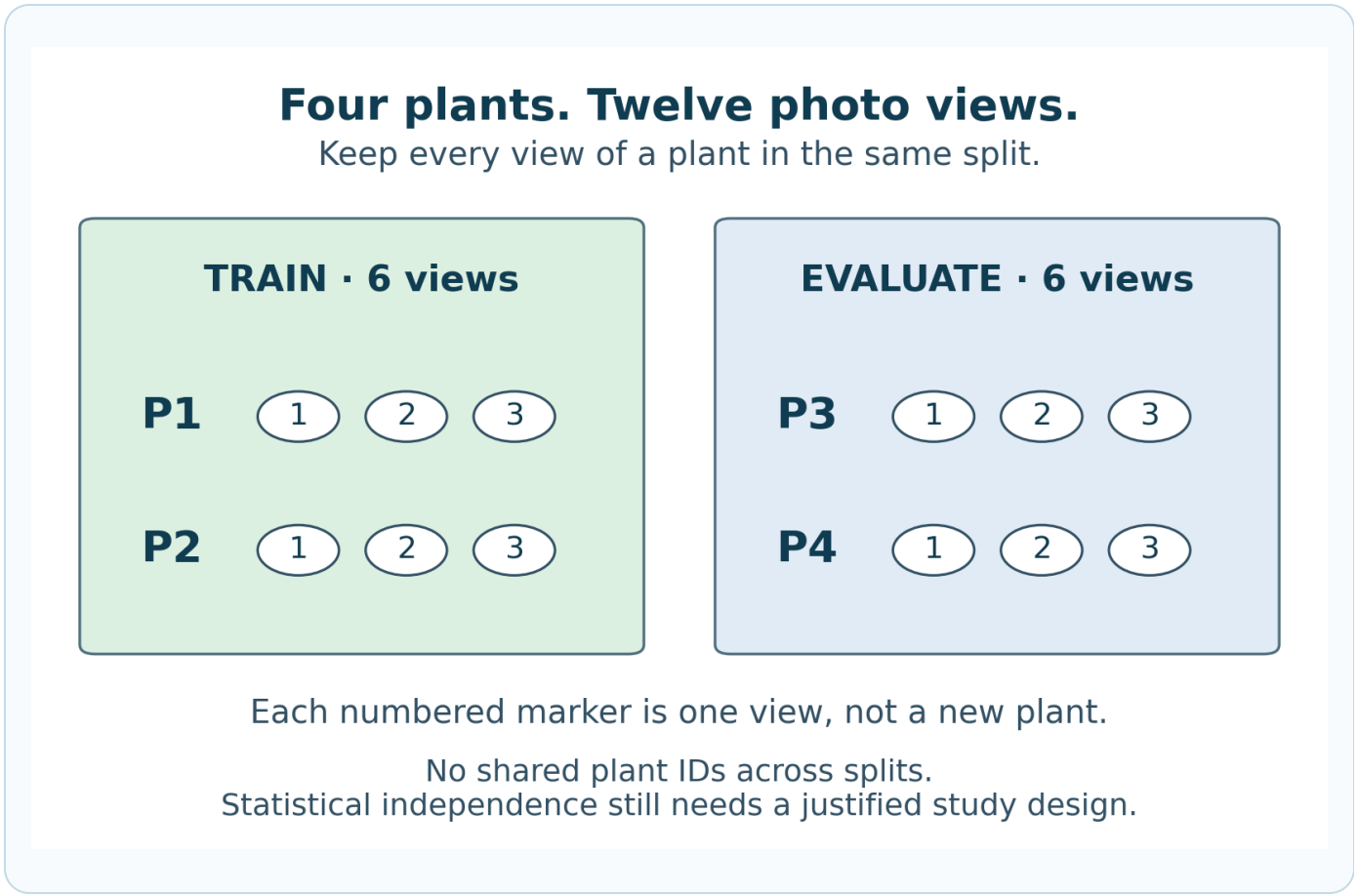
- Plant IDs P1–P4 each have three views. All views of a plant must stay in one split.

- Count distinct plant units and photo views, and state the independence assumption. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/9

Keep all views of a plant together

	Plant with 3 views	Plant with 3 views
Train: 6 photos	**P1**	**P2**
Evaluate: 6 photos	**P3**	**P4**

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/9

Experimental units and independence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify the unit

Four plants, not twelve independent plants, generated the observations.

2. Group before splitting

Assign whole plant IDs to training or evaluation.

3. Trace a valid split

Train on P1,P2 and evaluate on P3,P4: six photos per split.

CHANGE ONE THING

Put one P1 view into evaluation while other P1 views train.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.
- Which labelled step matches this detail? Train on P1,P2 and evaluate on P3,P4: six photos per split.
- 1. Group before splitting
- 2. Identify the unit
- 3. Trace a valid split
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Trace a valid split Train on P1,P2 and evaluate on P3,P4: six photos per split. Repeated views from one object share information.

Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

Changed-case question and optional hint

- Changed-case practice: Put one P1 view into evaluation while other P1 views train. Which conclusion is supported by the supplied facts?
- 1. Flag object-level leakage for this new-plant target.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Flag object-level leakage for this new-plant target. Evaluation now includes a plant already represented during **training**.

Repair and retain the record

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Experimental units and independence: the case

AI ACADEMY / AI-7/9 · 01 · LOOK AT THE FACTS

Experimental units and independence: the case

The situation

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

Supplied evidence

Plant IDs P1–P4 each have three views. All views of a plant must stay in one split.

Explain experimental units and independence using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

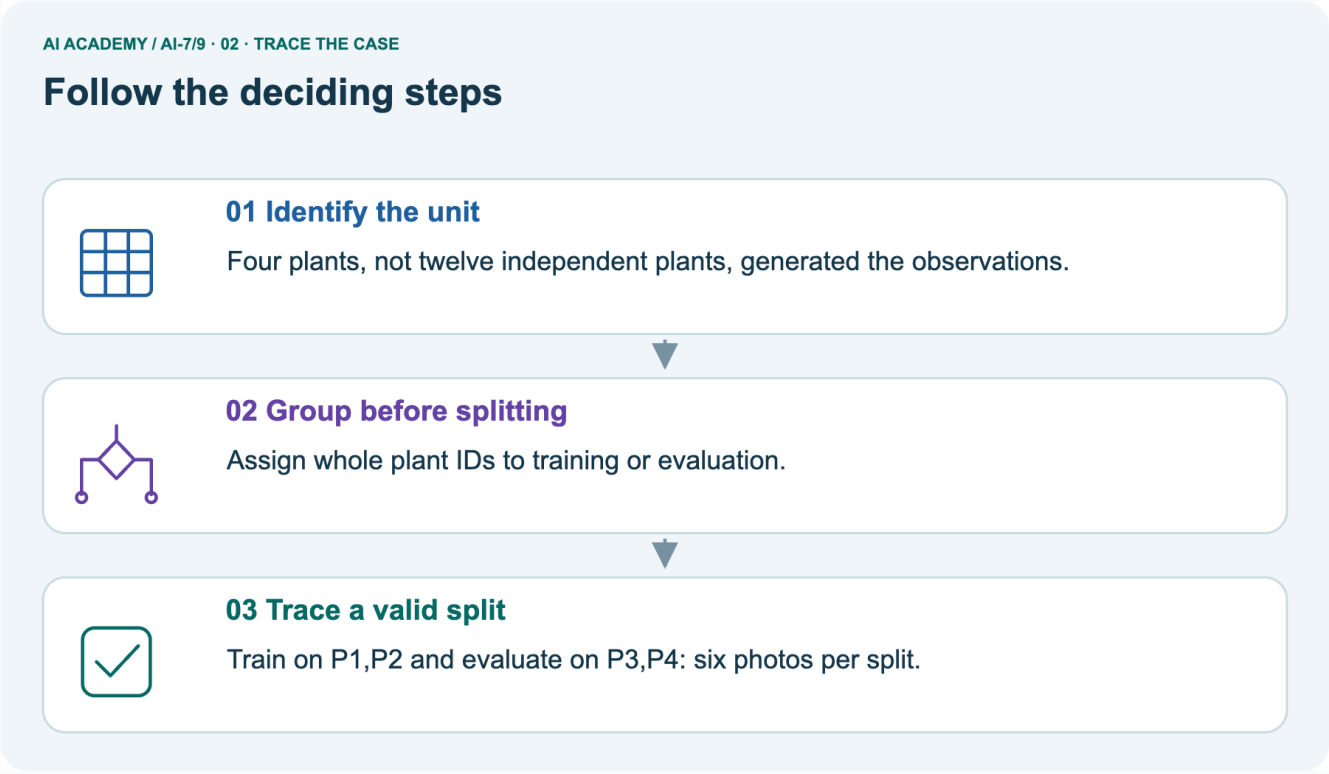
Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants. Supplied facts: Plant IDs P1–P4 each have three views.

All views of a plant must stay in one split.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



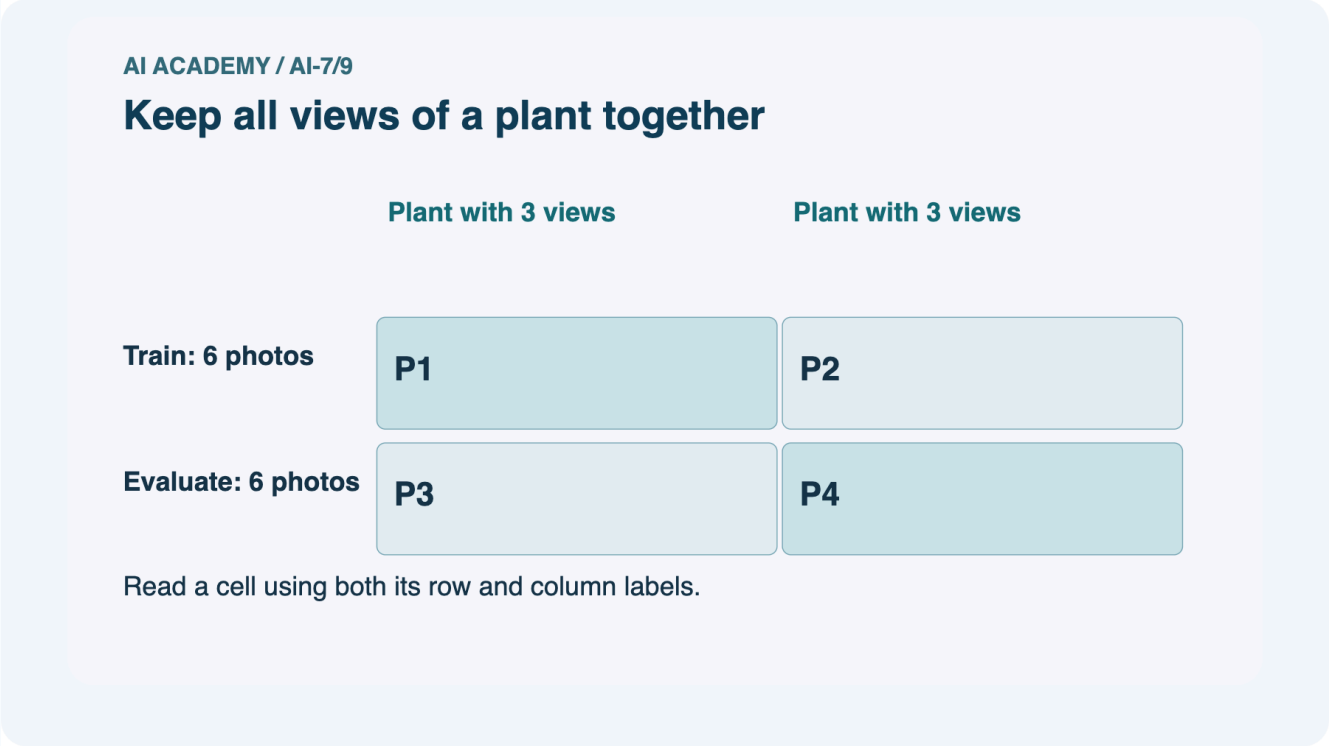
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the unit: Four plants, not twelve independent plants, generated the **observation**s. → Group before splitting: Assign whole plant IDs to **training** or evaluation. → Trace a valid split: Train on P1,P2 and evaluate on P3,P4: six photos per split.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep all views of a plant together



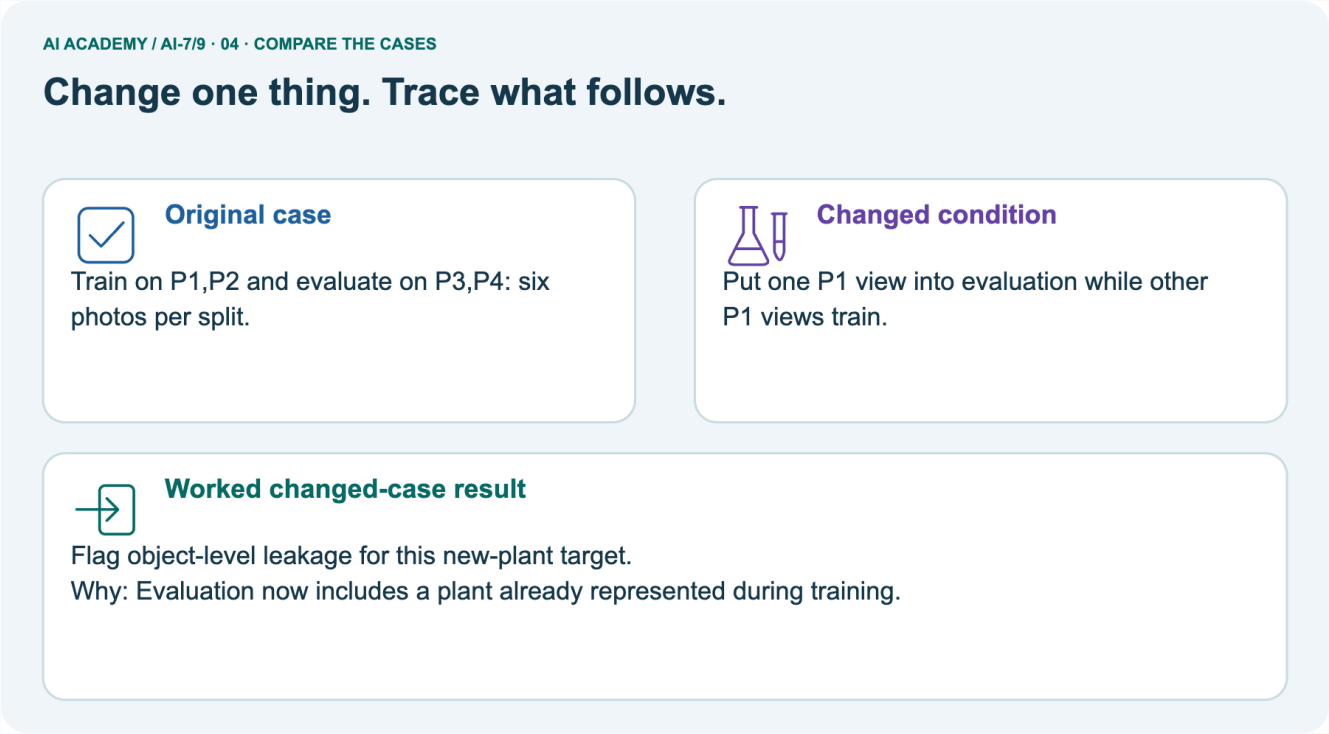
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep all views of a plant together. Train: 6 photos: Plant with 3 views = P1, Plant with 3 views = P2. Evaluate: 6 photos: Plant with 3 views = P3, Plant with 3 views = P4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

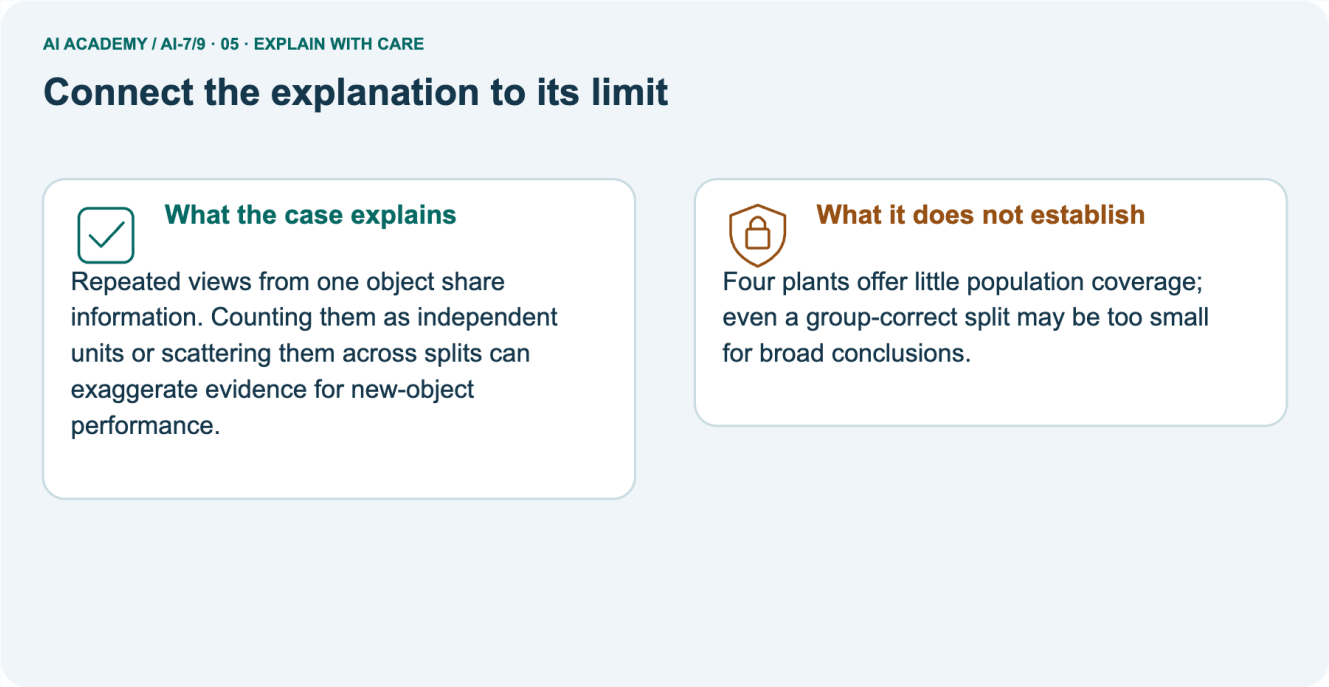
Original case: Train on P1,P2 and evaluate on P3,P4: six photos per split. Changed condition: Put one P1 view into evaluation while other P1 views train. Expected result: Flag object-level leakage for this new-plant target.

Reason: Evaluation now includes a plant already represented during **training**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

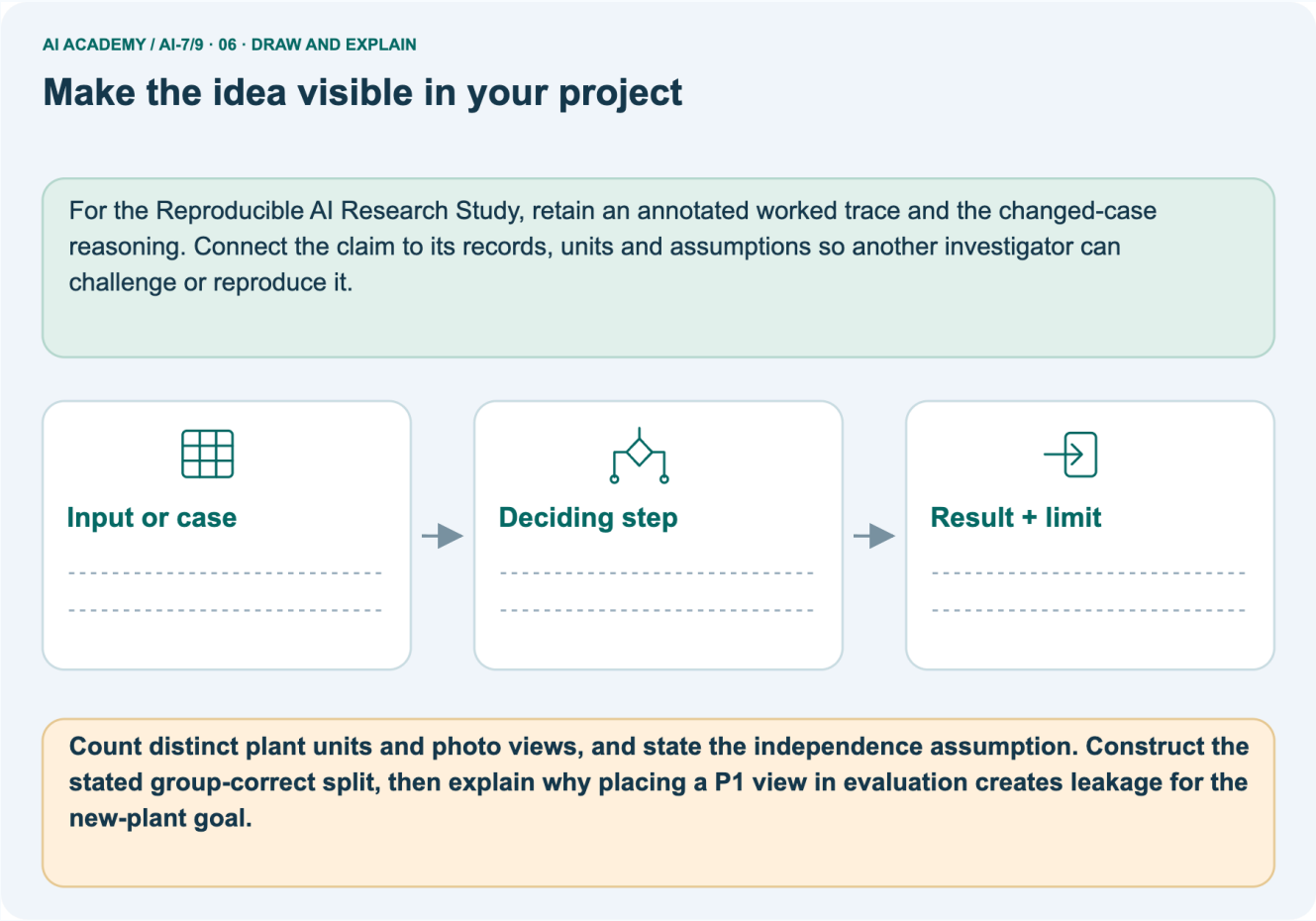
Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

Limit: Four plants offer little population coverage; even a group-correct split may be too small for broad conclusions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it. Draw your **input**, deciding step, result and limit. Supported question: Count distinct plant units and photo views, and state the independence assumption. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Experimental units and independence: the case

02. Follow the deciding steps

03. Keep all views of a plant together

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.